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Machine learning-enabled construction project management: systematic review, comparative performance synthesis and implementation framework

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Abstract

Purpose – Machine learning (ML), a subset of artificial intelligence (AI), is increasingly transforming construction project management (CPM) processes, yet a significant gap exists between research advancements and practical implementation. This study synthesizes empirical findings and proposes a framework to bridge the research-practice divide.

Design/methodology/approach – A systematic review was conducted using Scopus and Web of Science databases, following PRISMA guidelines. 64 peer-reviewed studies published between 2015 and 2025 were selected and coded using open and axial methods. The analysis was guided by systems theory, complexity theory, and dynamic capabilities theory to develop the Machine Learning–Enabled Construction Project Management (MLCPM) framework.

Findings – ML use in CPM aligns with three core areas: (1) planning tasks like cost estimation and scheduling, (2) execution tasks such as progress tracking and risk detection and (3) monitoring activities including cost control, delay prediction and performance tracking. The MLCPM framework offers a four-layer architecture consisting of data, ML processing, integration and output, with feedback and retraining loops to support scalable deployment. Furthermore, a quantitative synthesis of ten comparative studies revealed that hybrid and ensemble methods achieved superior performance in cost estimation (cost) (e.g. LightGBM and DNN-SVR), tree-based ensembles (random forest (RF) and gradient boosted trees (GBTs)) showed optimal accuracy for duration forecasting (duration) and hybrids with metaheuristic optimization outperformed single algorithms in delay prediction (delay).

Originality/value – This review consolidates existing literature on ML applications in CPM and introduces a comprehensive MLCPM Framework. Additionally, a quantitative subset synthesis of comparative evaluations extracts task–method performance patterns, enhancing decision relevance. We provide practical, theory-grounded guidance for adopting ML in construction.

Keywords Construction project management, Artificial intelligence, Machine learning, Implementation, MLCPM framework

Paper type Literature review

1. Introduction

Machine learning (ML), a subfield of artificial intelligence (AI), has increasingly transformed construction project management (CPM) by enabling data-driven forecasting, adaptive planning, and enhanced decision-making (Müller *et al.*, 2024; Taboada *et al.*, 2023). In recent years, ML has been applied to a wide range of CPM tasks, including cost estimation (Darko *et al.*, 2023), schedule optimization (Liben *et al.*, 2024), risk detection (Ma *et al.*, 2021), quality prediction (Fan, 2025), and safety analysis (Xia *et al.*, 2024).

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Despite the recognized potential of ML in project management (PM), a persistent gap remains between academic research and industrial implementation (Nenni *et al*., 2024; Shang *et al*., 2023; Zadeh *et al*., 2024). McKinsey (2020) reported that large construction projects typically take 20% longer to complete than scheduled and can be up to 80% over budget. These inefficiencies are often linked to the continued reliance on static, outdated tools–such as spreadsheets and slide decks–that are insufficient for managing modern construction projects' dynamic, multi-variable nature (Nieto-Rodriguez and Vargas, 2023). A key barrier to improvement is practitioners' limited familiarity with ML techniques, including challenges in selecting appropriate algorithms for specific tasks and interpreting model outputs (Mohammad and Chirchir, 2024). Moreover, the challenge includes evaluating these algorithms' effectiveness across diverse project types and specific processes (Zadeh *et al*., 2024).

Early studies established the value of ML in CPM. Kim *et al.* (2004) showed that neural networks outperformed traditional cost estimation methods, and Flood and Kartam (1994) proposed one of the earliest structured ML applications in civil engineering. Shenhar and Dvir (2007) called for adaptive, data-driven project systems–anticipating the current shift toward ML-integrated project environments.

While studies highlight the importance of integrating ML into PM processes–particularly in high-risk sectors like construction (Nenni *et al*., 2024; Shang *et al.*, 2023), a significant gap remains in translating these advancements into practical applications for real-world projects (Nenni *et al*., 2024). Existing reviews have largely addressed specific PM functions (Nenni *et al*., 2024; Taboada *et al*., 2023) or agile PM (Zadeh *et al*., 2024). Moreover, Tayefeh Hashemi *et al.* (2020) have primarily focused on cost prediction and estimation–without offering a comprehensive view of ML applications to other CPM functions or a theoretically grounded framework to contextualize these applications within CPM domains. Their study also lacks a conceptual framework, such as Systems or Dynamic Capabilities Theory, to structure the integration of these technologies across project lifecycles. This review addresses these gaps by (1) systematically mapping ML applications to core CPM processes, (2) evaluating their real-world effectiveness, and (3) providing a practical, theory-informed framework to guide implementation in the construction sector. Additionally, this review includes a quantitative synthesis of comparative studies to identify which ML approaches consistently outperform others in specific CPM tasks. Specifically, this review addresses the following research questions (RQs):

RQ1. How is ML implemented across core CPM processes, and what patterns, tools, and techniques are most applied?

RQ2. How effective are ML models in construction PM applications, and what challenges or limitations impact their performance?

In this study, AI refers to systems performing tasks requiring human-like reasoning, learning, and decision-making (Russell and Norvig, 2021). As a subset of AI, ML uses data-driven algorithms that improve without explicit programming. PM involves the structured use of knowledge, skills, tools, and techniques across phases such as planning, execution, and monitoring, following the Project Management Body of Knowledge (PMBOK) framework (PMI, 2021). These definitions guided our search and inclusion criteria, aligning theoretical scope with construction-specific applications.

2. Theoretical foundation

To guide framework development, theories were selected for their relevance in addressing ML integration challenges in PM, based on three criteria: (1) offering a comprehensive lens to handle complexity, uncertainty, and adaptability (Turner and Müller, 2005; Geraldi *et al.*, 2011); (2) integrating theoretical and empirical insights to build a cohesive AI-enhanced PM framework (Richard *et al*., 2009); and (3) supporting advanced decision-making, such as

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resource allocation, risk mitigation, and system optimization (Marnewick, 2016). After evaluating several options, Systems Theory, Complexity Theory, and Dynamic Capabilities Theory were selected as core theories for the MLCPM framework, preferred over alternatives like Actor-Network Theory or Technology Acceptance Model, which lack the necessary comprehensive scope (Sheffield *et al.*, 2012; Bakhshi *et al.*, 2016; Davies and Brady, 2016). These theories have proven especially valuable in technology-enabled construction PM and digitalization efforts (Hosseini *et al.*, 2018; Wang *et al.*, 2024).

2.1 Systems theory

Rooted in interconnectedness (Bertalanffy, 1969; Sterman, 2002), Systems Theory offers a lens to manage PM complexity through feedback loops, nonlinear dynamics, and interdependent structures. It views projects as open systems exchanging resources and information with their environments (Ackoff and Gharajedaghi, 1996). Key principles–holism, boundary definitions, feedback loops, and emergence–are vital for understanding how project elements interact (Zhu and Mostafavi, 2017).

Construction PM has been used to model stakeholder interactions, information flows, and performance metrics (Locatelli *et al.*, 2014). Combined with AI/ML, Systems Theory enables integrated data analysis and decision-making for risk mitigation, resource allocation, and progress tracking.

2.2 Complexity theory

Complexity Theory addresses the unpredictability of projects influenced by technical, organizational, and environmental factors, viewing them as complex adaptive systems with non-linearity, emergence, and sensitivity to initial conditions (Cicmil *et al.*, 2006; Bakhshi *et al.*, 2016). Being unique and multi-organizational, construction projects are especially suited to this lens (Cicmil *et al.*, 2017).

The theory explains why minor input variations can cause significant output shifts–common in cost and schedule deviations (Qazi *et al.*, 2016). ML applications support adaptive algorithms that detect complex patterns, outperforming linear models by capturing nonlinear relationships (Maylor and Turner, 2017; Wood and Ashton, 2010).

2.3 Dynamic capabilities theory

This theory focuses on how organizations sustain competitive advantage in dynamic settings by sensing opportunities, seizing them, and transforming resources accordingly (Teece, 2007; Teece *et al.*, 2016). Unlike static resource-based views, it emphasizes routines for adapting to evolving technologies.

Its three core processes–sensing, seizing, and transforming (Davies and Brady, 2016) –are essential for construction firms adopting ML, as they must align new technologies with operational workflows. It also explains why technically sound AI solutions fail without organizational readiness (Gajendran *et al.*, 2014; Wong *et al.*, 2008).

These three theories provide a multi-dimensional foundation for understanding and addressing the complexities of ML integration in CPM.

3. Methodology

3.1 Data collection and screening

To identify relevant studies, we searched two major research databases, Scopus and Web of Science Core Collection, covering the period from January 2015 to April 2025. The timeframe (2015–2025) captures ML's practical shift in construction project management, aligns with PMBOK 6th Edition, and reflects recent technological advancements. The search was designed to capture ML applications in CPM, with keywords including *"machine learning," "ML,", "AI", "artificial intelligence", "deep learning,"* and *"project management"*.

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The search yielded 1,179 results from Scopus and 486 from Web of Science. After removing duplicates using Excel filtering, 1,336 records remained for screening. This study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework (Page *et al*., 2021) to guide the literature review process. PRISMA provides a transparent and structured approach to identify, screen, and synthesize relevant studies, ensuring scientific rigor and reproducibility. We applied the following inclusion criteria:

(1) Peer-reviewed empirical articles in English.

(2) Focused on ML implementation in CPM, aligned with PMBOK (6th edition) process groups (e.g. planning, execution, monitoring, and control).

(3) Published between 2015 and April 2025.

(4) Demonstrated ML use in real-world or dataset-driven CPM tasks with measurable outputs.

Articles were excluded if they were.

(1) Non-empirical works (e.g. reviews, editorials, theoretical papers).

(2) Studies outside construction or general PM without construction context.

(3) Articles published before 2015 or in non-English languages.

To minimize the risk of exclusion, we included ambiguous cases for full-text review. Of the 234 papers, 172 were excluded after a qualitative assessment for not addressing the review's questions, and 2 studies were added from the references of included papers. The final sample consisted of 64 papers from 42 journals (Figure 1).

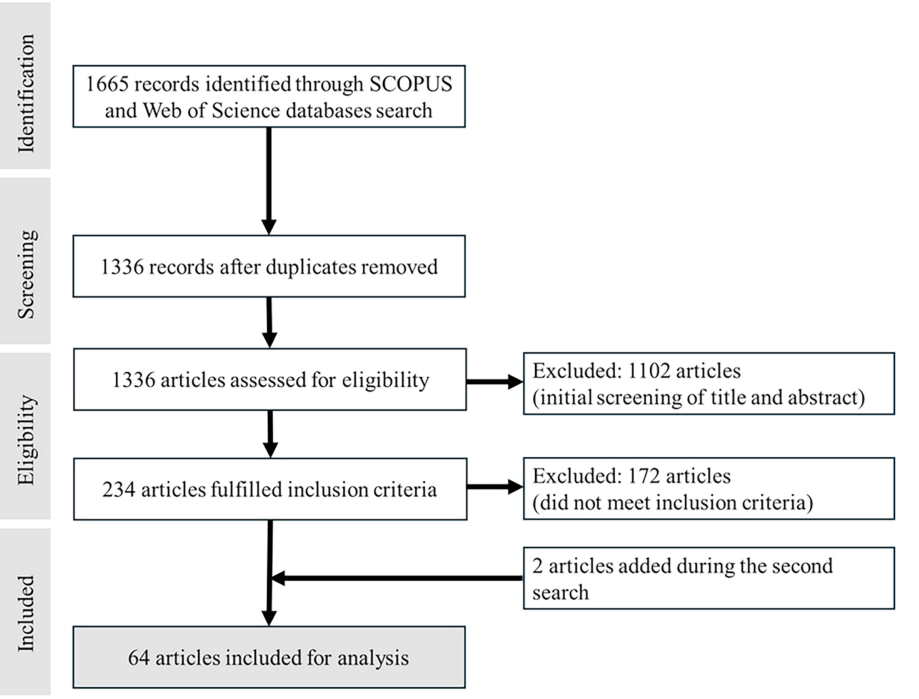


Figure 1. Flow chart of the different phases of the systematic review. Source: Authors' own work

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3.2 Data extraction, coding, and theoretical integration

After finalizing the 64 articles, we conducted a structured data extraction process to capture key information on the ML techniques, targeted CPM functions, performance metrics, and implementation challenges. Each article was examined for alignment with PMBOK process groups (planning, execution, monitoring, and control) and categorized based on the ML methods applied (e.g. neural networks, ensemble models, fuzzy systems, graph-based learning). The analysis followed a three-stage coding procedure to ensure analytical rigor. In the first stage, open coding identified recurring ML application areas such as cost estimation, delay prediction, risk analysis, and schedule optimization. These initial codes were then grouped through axial coding into 12 subthemes, including resource allocation, progress monitoring, and performance tracking. Finally, in the selective coding stage, these subthemes were synthesized into three main themes aligned with construction project phases: planning (29 studies), execution (15 studies), and monitoring and control (20 studies). Table 1 summarizes the 64 studies by phase, ML methods, performance metrics, and implementation factors.

Figure 2 summarizes the 64 studies. Figure 2(a) maps ML method families by project phase, with Planning most represented and hybrids and tree-based models prevalent across phases. Figure 2(b) charts annual publications from 2016 to 2025, rising after 2020, peaking in 2024, and continuing in 2025.

We employed coding procedures informed by grounded theory and thematic analysis (Strauss and Corbin, 1990). Open coding involved line-by-line analysis identifying ML methods, data types, and performance metrics. Axial coding connected these concepts (e.g. linking algorithms to CPM phases). Selective coding integrated codes into core themes aligned with our theoretical framework.

Two independent reviewers manually coded the dataset using Microsoft Excel. Inter-rater reliability was assessed using Cohen's kappa (κ), which measures agreement between two raters for categorical items while accounting for chance agreement (McHugh, 2012). κ ranged from 0.61 to 1.0, indicating substantial to near-perfect agreement. Any disagreements during coding or theme assignment were resolved through discussion; a third expert reviewed unresolved cases to reach a consensus. To support analytical depth and coherence, we integrated three guiding theoretical frameworks into our methodology: Systems Theory, Complexity Theory, and Dynamic Capabilities Theory. Systems Theory (Bertalanffy, 1969; Sterman, 2002) shaped our interpretation of interconnected project functions and iterative feedback processes, particularly in cases where ML models supported system-wide decision-making. Complexity Theory (Cicmil *et al.*, 2006) helped frame our evaluation of ML model types–especially hybrid and ensemble approaches–as responses to non-linearity, uncertainty, and emergent project behaviors. Meanwhile, Dynamic Capabilities Theory (Teece, 2007) focused on organizational readiness, enabling us to capture data governance, tool integration, retraining protocols, and human-machine interaction themes. Together, these theoretical lenses provided a multi-dimensional foundation for analyzing the technical performance of ML models and their contextual fit, organizational adaptability, and alignment with dynamic construction environments.

The 64 papers span 42 journals with publication activity concentrated between 2022 and 2025. Our methodology was guided by an integrated framework combining Systems, Complexity, and Dynamic Capabilities Theory. Systems Theory (Bertalanffy, 1969; Sterman, 2002) informed our data extraction by highlighting the interconnectedness of PM processes. This issue led to a coding structure that captured the relationships between AI/ML applications and their impacts across project phases, emphasizing feedback mechanisms and system boundaries over isolated analysis.

Complexity Theory (Cicmil *et al.*, 2006) guided our examination of how AI/ML technologies address nonlinear dynamics and uncertainty in construction projects. We categorized studies by their ensemble, hybrid, or graph-based modeling approaches and assessed their ability to capture emergent project behaviors.

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Table 1. Summary of the 64 articles reviewed in this study

Application phase	ML method(s)	Model performance	Success factors	Reference
PLANNING PHASE	Hybrid ML + ANN	Accuracy 97.42%	Feature selection, expert validation	Deepa *et al.* (2025)
	Node2vec, Word2vec, Cosine Similarity	82% accuracy, 40% precision	Graph representations, contingency planning	Mostofi *et al.* (2025b)
	XGBoost-SA	92% accuracy, $R^2 = 0.699$	Hybrid optimization, ensemble learning	ForouzeshNejad *et al.* (2024)
	Graph Convolutional Network	Accuracy 85%, F1 score 78–80%	Graph modeling, rework cost prediction	Mostofi *et al.* (2024b)
	DBO + BPNN, GA + BPNN, PSO + BPNN	$R^2 = 0.9884$, MAPE = 17.56%	Optimization algorithms, bootstrapping	Wang *et al.* (2024)
	SSAE	MAPE 0.71%, RMSE 675,842	Historical datasets, feature reduction	Zhou *et al.* (2024)
	LSMBM (LSSVM + BPNN + PCA)	RMSE 0.016, accuracy 95.43%	PCA integration, hybrid models	Cheng and Khasani (2024)
	GRA, LASSO, BPNN	MAE 197.02, RMSE 234.64	Gray relational analysis, domain expertise	Chen and Wang (2024)
	RF, NN, LR, DT	93% PPC correlation	Last Planner System, performance forecasting	Ezzeddine *et al.* (2022)
	LightGBM, Ensemble	MAPE 2.99%, $R^2 = 0.931$	SHAP visualization, expert collaboration	Elmousalami *et al.* (2025)
	RF, KNN, SVM, CART, NN	RF: RMSE 74 days, R^2 0.97	Ensemble techniques, categorical data	Liben *et al.* (2024)
	XGBoost, CatBoost, SVM	$R^2 = 0.985$ (stacked model)	Stacking ensemble, feature engineering	Ahn *et al.* (2024)
	SVM, DT, Gradient Boosting	$R^2 = 0.88$, ACC = 91%	Bootstrap sampling, steel fabrication focus	Shahedi *et al.* (2024)
	Node2vec, Word2vec, NN	Accuracy 94–96%, AUC 97%	Graph representations, resource optimization	Mostofi *et al.* (2024a)
	Improved AO, Deep Auto-Coder	CPI improved to 1.079	Optimization integration, manpower planning	Ding *et al.* (2024)
	Multinomial Naïve Bayes	Qualitative validation confirmed	Familiar tool integration, expert input	Assaf and Assaad (2024)
	DNN, SVR	MAPE: 0.06 (cost), 0.07 (duration)	Deep learning, green building focus	Darko *et al.* (2023)
	CNN, NB, SVM, LR, RF	82.72% precision, 87.38% recall	CNN integration, requirement prioritization	ul Hassan *et al.* (2023)
	BP NN, Entropy	97% accuracy	Neural-entropy hybrid, provider selection	Bai *et al.* (2022)
	KNN	Accuracy 86%, Recall (under-budget) 0.99, Precision 0.86	Simplicity, procurement method as key feature, data preprocessing	Kusonkhum *et al.* (2022)
	SVR, Monte Carlo	Improved buffer allocation	Simulation integration, critical chain planning	Li *et al.* (2022)
	SDAE	Average error <1.54%, R^2 0.9743	Stacked denoising, foundation pit focus	Liu *et al.* (2020)
	LSTM, GRU	MAE as low as 0.26%; 7-step-ahead forecasting	Hyperparameter tuning, real project data, model comparison	Liu and Hao (2021)
	DBM-SoftMax, BPNN, SVM	MSE ~0.022 (DBM-SVM)	Economic variables, hybrid deep learning	Rafiei and Adeli (2018)
	ANN (MLP)	R^2 0.973, accuracy 93.5%	Historical datasets, dam construction	Golizadeh *et al.* (2017)
	MLP, RBF	RBF: $R^2 = 0.9871$	Comparative analysis, early-stage estimation	Bayram *et al.* (2016)
	KNN, SVR, GBT, ANN	GBT: RMSE 2.207, R^2 0.961	Gradient boosting, time-series preprocessing	Zhang and Li (2024)
	RL, Graph-based	Automated adaptive scheduling	Reinforcement learning, constraint handling	Yao *et al.* (2024)
	ANN, EVM, BN, Hybrid EVM-ANN	Hybrid: 94.46% accuracy	EVM integration, hybrid scheduling	Soetjipto *et al.* (2024)

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Table 1. Continued

Application phase	ML method(s)	Model performance	Success factors	Reference
EXECUTION PHASE	CNN, CSO, NSGA-II	Accuracy 97.56%, Precision 0.97	Optimized CNN, prefabricated construction	Zeng and Gao (2025)
	BERT, RNN, SVM, Ensemble	Accuracy: 89%, F1: 86%	NLP integration, contract analysis	Dikmen *et al.* (2025)
	Bayesian Network, GAN	Accuracy 72%–85% post-GAN	GAN augmentation, risk identification	Khodabakhshian *et al.* (2025)
	DEMATEL, ANP, ANN	Spearman 1.0	Multi-criteria integration, environmental risk	Ershadi and Ashtiyani (2019)
	ANN (Multilayer Perceptron)	MSE 0.0029, R^2 0.966	Multi-risk modeling, cyber-physical risks	Aggabou *et al.* (2024)
	ANN, Transformers	Accuracy: 90.2%, Loss: 0.00025	Transformer architecture, contract risk	Hamdy *et al.* (2024)
	RF, XGBoost, NGBoost	RFR: RMSE 0.0745, MAPE 0.396	SHAP interpretability, safety cost prediction	Xia *et al.* (2024)
	DT, KNN	76% accuracy	Decision tree transparency, safety performance	Abbasianjahromi and Aghakarimi (2023)
	CNN, Image Processing	Accuracy 81%, Recall 83%	Computer vision, progress monitoring	Greeshma and Edayadiyil (2022)
	CNN, RF, Apriori	99% accuracy	Cascading effects, safety risk analysis	Ma *et al.* (2021)
	DT, RF, XGBoost, LSTM	>90% maintenance accuracy	Platform integration, equipment prediction	Choi *et al.* (2021)
	DNN, MRA	41.3% lower MAE	Deep learning, financial loss prediction	Kim *et al.* (2022)
	ANN	MAPE low, APD 0.08	Behavioral modeling, safety climate	Patel and Jha (2016)
	Bid-change analysis	Uncertainty quantification	Historical data comparison, risk forecasting	Shrestha *et al.* (2025)
	BBN	80% accuracy	Bayesian networks, complexity measurement	Luo *et al.* (2020)
Monitoring and control	Deep Learning, NN, DT, NB	DL accuracy 90.79%	Deep architecture delay cause prediction	Ouansrimeang and Wisaeang (2024)
	ANN, ANFIS, GPR, LSTM, SVM	ANN: MAPE 0.2376, R^2 0.9878	Fuzzy logic integration, EVM metrics	Yalçın *et al.* (2024)
	CatBoost, XGBoost, LGBM	LGBM: accuracy 0.85, precision 0.80	Regional adaptation, delay risk classification	Alsulamy (2025)
	Fuzzy Deep Learning	>96% accuracy	Fuzzy-weighted features, quality prediction	Fan (2025)
	SULSTM, Multi-objective	Multi-objective optimization	Cost-quality-duration trade-offs	Zhou and Guo (2024)
	RFC	54–81% accuracy range	Random Forest, success factor analysis	Bang *et al.* (2022)
	MLP-GA, MLP-PSO	RMSE 5.31 months, R^2 0.9916	Metaheuristic optimization, time/cost prediction	Bakhshi *et al.* (2022)
	ANFIS, MLP, RBF, GRNN	ANFIS: MSE 0.0003, RMSE 0.0181, R = 0.996	Neuro-fuzzy integration, model optimization, sensitivity analysis	Dastgheib *et al.* (2022)

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Table 1. Continued

Application phase	ML method(s)	Model performance	Success factors	Reference
	ANN	84.21% accuracy	Building performance, schedule/cost prediction	Juan and Liou (2022)
	ANN	Accuracy 93.75%, Kappa = 0.89, Precision = 0.96, FP Rate = 0.013	Expert-driven feature selection, dimensionality reduction, systematic hyperparameter tuning	Sanni-Anibire *et al.* (2022)
	Ensemble (Bagging, Boosting, NB)	ROC AUC >0.74	Ensemble diversity, delay prediction	Egwim *et al.* (2021)
	Fuzzy EVM, BP Neural Network	32.06% MAPE improvement in schedule prediction	Hybrid model design, fuzzy logic, proactive control, explainability	Yu *et al.* (2021)
	RF-GA	91.67% accuracy	Genetic optimization, delay risk prediction	Yaseen *et al.* (2020)
	ANN	65.5% time, 65.4% cost accuracy	Post-disaster adaptation, deviation prediction	Waheeb *et al.* (2020)
	DT, NB	NB: 78.4% train accuracy	Interpretable models, delay risk prediction	Gondia *et al.* (2020)
	ANN	97% reliability	Highway focus, schedule performance prediction	Jasim *et al.* (2020)
	Multihead GAT	Accuracy: 76.8%, AUC 0.78	Attention mechanisms, delay prediction	Mostofi *et al.* (2025a)
	RBF Neural Network, RCF	R^2 = 0.991 (train), 0.964 (test)	Reference class forecasting, resilience modeling	Chen *et al.* (2024)
	MLP, CROA	R^2 0.9914, MAE 7.07%	Optimization integration, time overrun prediction	Shihadeh *et al.* (2024)
	SVM, RF, k-NN, XGBoost	76–90% accuracy range	Network analytics, performance prediction	Uddin *et al.* (2024)

Note(s): Abbreviations: EAC = Estimate at Completion; PPC = Percent Plan Complete; RCF = Reference Class Forecasting; APD = Absolute Percentage Difference; BBN = Bayesian Belief Network; RFC/RFR = Random Forest Classifier/Regressor; DBO = dandelion-based optimizer; CSO = cat swarm optimization; CROA = coral reef optimization algorithm; AO = arithmetic optimization

Source(s): Authors' own work

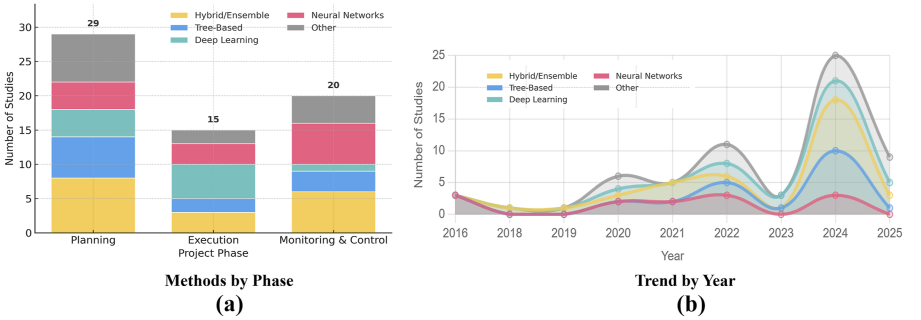


Figure 2. ML methods by project phase and publication trend (2016–2025, n = 64). Source: Authors' own work

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Dynamic Capabilities Theory (Teece, 2007) directed our focus toward organizational enablers and barriers to AI/ML adoption, coding for capabilities related to sensing opportunities, seizing them through implementation, and transforming routines to support ongoing adaptation.

3.3 Quantitative synthesis of comparative model performance

To quantify patterns in model performance, we conducted a rank-based synthesis of model performance patterns from studies that employed comparative evaluations on identical datasets. Studies were included if they (1) evaluated two or more ML models on the same task and (2) reported quantifiable performance metrics. This criterion yielded 10 eligible studies out of 64.

Notation list:

Sets:

s:	Study index; k: number of models compared in study, $s \in \{1, \ldots, \overline{s}\}$
a:	Model index
A:	Algorithm family (e.g. ANN, Tree/Boosting, Hybrid)
A_s:	Set of models in family A evaluated in study s
T:	Task (Cost, Duration, Delay)
S_T:	Set of eligible studies for task T,

Parameters:

$\mid S_T \mid$:	the number of eligible studies for task T
w_s:	Non-negative study weight

Decision variables:

$\text{Rank}_{a,s}$:	Within-study rank (one is best rank), $\text{Rank}_{a,s} \in \{1, \ldots, k\}$, $k \in \mathbb{N}$, $k \geq 2$
$R_{a,s}$:	Normalized score of model a in s, $R_{a,s} \in [0, 1]$
$\overline{R}_{A,s}$:	Mean of $R_{a,s}$ over $a \in A_s$ (Family score inside study s)
$\text{Score}_{A,T}$:	Pooled, evidence-weighted score for family A on task T

Rank-based scoring: For a study s with k models, we assign each model a rank-normalized score:

$$R_{a,s} = \frac{k - \text{Rank}_{a,s}}{k - 1} \tag{1}$$

Winner-only fallback: Used when a paper reports only the top model or mixes incomparable scenarios:

$$R_{a,s} = \begin{cases} 1, & \text{Rank}_{a,s} = 1 \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

Family score inside a study (absent families contribute zero):

$$\overline{R}_{A,s} = \begin{cases} \dfrac{1}{\mid A_s \mid} \sum_{a \in A_s} R_{a,s}, \mid A_s \mid \; > 0 \\ 0, \mid A_s \mid \; = 0 \end{cases} \tag{3}$$

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Task-level pooling across studies (Borda-type aggregation):

$$\text{Score}_{A,T} = \frac{\sum_{s \in \mathbf{S}_T} w_s \, \overline{R}_{A,s}}{\sum_{s \in \mathbf{S}_T} w_s} \tag{4}$$

By "Borda-type" we mean that studies contribute relative ranks (mapped to $[0, 1]$) rather than raw errors, and we then take the study-weighted mean across studies (w_s defined in Eq. (5)).

Study weighting. Weights combine validation rigor (α_s), sample size/type (β_s), and leakage risk (γ_s), multiplicatively, then are normalized to mean 1 within each task:

$$w_s^* = \alpha_s \beta_s (1 - \gamma_s), \; w_s = \frac{w_s^*}{\left(\frac{1}{|\mathbf{S}_T|} \sum_{u \in \mathbf{S}_T} w_u^* \right)} \tag{5}$$

Validation rigor (α_s): external/chronological hold-out = 1.00; repeated k-fold CV = 0.90; single k-fold CV/hold-out = 0.85; resubstitution/unclear split = 0.60.

(1) Size/type (β_s): large independent $N \geq 1000 = 1.00$; medium $100 \leq N < 1000 = 0.90$; small $N < 100 = 0.80$; single-project time-series = 0.70.

(2) Leakage penalty (γ_s): clear pre-split target leakage (or strong risk) = 0.10; otherwise = 0.00.

By construction, $\frac{1}{|\mathbf{S}_T|} \sum_{s \in \mathbf{S}_T} w_s = 1$.

Note: Weight values reflect established validation hierarchy in ML research. External/chronological hold-out (1.00) provides the strongest evidence of generalization. Repeated k-fold CV (0.90) reduces variance and selection bias compared to single k-fold/hold-out (0.85). Resubstitution (0.60) carries high overfitting risk. For sample size, larger independent datasets ($N \geq 1{,}000$) yield more reliable estimates (1.00), with graduated penalties for smaller samples (0.90 for N = 100–999; 0.80 for $N < 100$). Single-project time-series (0.70) are discounted for limited generalizability and temporal autocorrelation. The leakage penalty ($\gamma_s = 0.10$) dampens known optimistic bias from data contamination.

Table 2 reports these normalized w_s values. When studies report multiple performance metrics, we prioritize the authors' stated primary key performance indicator (KPI); otherwise, we default to mean absolute percentage error (MAPE) for cost, root mean squared error (RMSE) or mean absolute error (MAE) for duration, and accuracy/F1 for delay. Conflicting KPIs are rank averaged within studies before applying Eq. (3). Robustness checks confirm that ±10% weight perturbation and alternative KPI prioritization do not alter the main conclusions.

This rank-based aggregation approach (following Borda count principles) addresses the challenge of comparing disparate performance metrics (e.g. MAPE versus RMSE versus classification accuracy) while maintaining within-study experimental validity. We adhered to established best practices for external validation and acknowledged the interpretive limitations of percentage-based error metrics (Hyndman and Koehler, 2006; Vetterrotti and Di Camillo, 2021).

4. Findings

4.1 ML implementation in CPM (RQ1)

Across the 64 studies reviewed, ML applications are concentrated in planning, execution, and monitoring and control, showing increasing alignment with PMBOK process groups and a shift toward ensemble and hybrid models for complex project environments, with most implementations focusing on data-rich tasks in early and middle phases.

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Table 2. Study weights and best-performing models

Study (year)	Task	k (models)	Models/families compared	*N*/type	Validation	Normalized weight (w_s)	Best performance (metric)	Best-performing family
Elmousalami *et al.* (2025)[a]	COST*	NR (winner-only)	Tree ensembles (LightGBM, ET, RF) + baselines	1,277 (projects)	70/15/15 hold-out	1.16	MAPE: 2.99% (LightGBM)	TREE (LightGBM)
Ahn *et al.* (2024)[a]	COST	≥3	CatBoost, XGBoost (Tree), SVM; stacked CatBoost + SVM	35,194/517 (projects)	Time-split external	1.37	R^2: 0.985, MAE: 605.1k KRW (stacked)	HYBRID (CatBoost + SVM)
Darko *et al.* (2023)	COST	2	DNN (ANN), SVR; DNN–SVR	198 (projects)	Validation (no test split)	0.74	MAPE: 6.0% (cost, DNN–SVR)	HYBRID (DNN–SVR)
Chen and Wang (2024)	COST	3	ANN, LASSO; GRA–LASSO–BPNN	47 (projects)	70/30 hold-out	0.93	MAE: 197.02, RMSE: 234.64	HYBRID (GRA–LASSO–BPNN)
Bayram *et al.* (2016)	COST	2	ANN (MLP, RBF) + Statistical baseline	232 (projects)	Time-split external	1.23	MAE: 38,475, RMSE: 65,969; R^2: 0.987	ANN (RBF)
Yalçın *et al.* (2024)	COST	6	ANN, ANFIS, GPR, LSTM, SVM, M5Tree	Single-project TS	Resubstitution	0.57	MAPE: 23.76% (EAC-PV1, ANN)	ANN
Liben *et al.* (2024)	DURATION	5	RF (Tree), KNN, SVM, CART/DT, ANN	~100 (projects)	10 × 3 CV	1.03	RMSE: 74 days; R^2: 0.97 (RF)	TREE (RF)
Zhang and Li (2024) **	DURATION	4	GBT (Tree), ANN, SVM, KNN	722 rec. (one project)	Hold-out	0.97	GBT: MAE 2.39; RMSE 4.09 · ANN: RMSE 3.80	TREE (GBT)
Shihadeh *et al.* (2024)	DELAY	≥2	ANN vs CROA–MLP (Hybrid)	190 (projects)	Hold-out + CV	1.09	R^2: 0.9914; RMSE: 0.107 (CROA–MLP)	HYBRID (CROA–MLP)
Yaseen *et al.* (2020)	DELAY	≥2	RF vs RF–GA (Hybrid)	30 (projects)	k-fold CV	0.91	Accuracy: 91.67% (RF–GA)	HYBRID (RF–GA)

Note(s): Weights are normalized to mean 1 within task; values > 1 indicate above-average evidence weight, [a]Winner-only rows were scored using the Eq. (2) and are marked [a]
* Task for Elmousalami *et al.* (2025) study counted under cost only (duration-variance excluded by design)
** Sensitivity (Zhang and Li, 2024): ANN is best by RMSE (3.80), but the authors prioritize GBT (slightly higher RMSE, much faster), so we keep GBT as the within-study winner
Weights computed via Eq. (5) using α_s (validation rigor), β_s (sample size/type), and γ_s (leakage penalty) as defined in Section 3.3. Weights are task-local; cross-task comparisons of w_s are not meaningful
Abbreviations: NR = not reported; CV = cross-validation
Task distribution: Cost (n = 6), Duration (n = 2), Delay (n = 2)
Source(s): Authors' own work

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4.1.1 Planning phase. ML is predominantly applied to cost estimation, schedule optimization, and resource allocation in the planning phase. Cost estimation is the most mature area, where ensembles and hybrids such as Extreme Gradient Boosting XGBoost, LightGBM, and deep neural networks (DNN) achieve high accuracy (Elmousalami *et al*., 2025; Wang *et al*., 2024). Studies report R squared values above 0.95 and report contingency reserves moving from 10–15% to 5–8% (Liu *et al*., 2020; Dastgheib *et al*., 2022). Feature selection and interpretability tools such as SHAP, LASSO, PCA, and gray relational analysis improve robustness and transparency (Chen and Wang, 2024; Cheng and Khasani, 2024).

Schedule optimization frequently employs artificial neural networks (ANNs), sometimes with earned value methods or with support vector regression (SVR). Tree based methods and recurrent models such as long short-term memory (LSTM) and gated recurrent unit (GRU) support multi step forecasting of resource linked variables (Liben *et al*., 2024; Liu and Hao, 2021). Reinforcement learning appears for adaptive scheduling but remains largely in simulation (Yao *et al*., 2024).

Resource allocation evolves from simple classifiers to graph informed decision support. Examples include node2vec for relational modeling and neural optimization for labor planning, indicating growing context awareness (Mostofi *et al*., 2024a; Ding *et al*., 2024).

4.1.2 Execution phase. In the execution phase, ML applications focus on progress monitoring, risk detection, and safety management. Progress monitoring increasingly uses convolutional neural networks (CNNs) based computer vision with streaming pipelines, achieving 81–97% accuracy in masonry progress and prefab module placement (Greeshma and Edayadiyil, 2022; Zeng and Gao, 2025). These tools support site reporting, quality control, and automated delay flags.

Risk detection shifts from qualitative judgment to proactive ML. Transformer based NLP and Bayesian networks surface legal, contractual, and operational risks with reported accuracy above 84% (Dikmen *et al*., 2025; Khodabakhshian *et al*., 2025). Integrated platforms such as EMAP combine multiple methods for continuous tracking (Choi *et al*., 2021). Safety studies range from early ANN predictors to recent ensembles that estimate cost and severity of incidents. DNNs that incorporate weather and environmental variables provide prescriptive guidance, and interpretable models help prioritize interventions (Kim *et al*., 2022; Abbasianjahromi and Aghakarimi, 2023; Xia *et al*., 2024).

4.1.3 Monitoring and control phase. Monitoring and control applications focus on cost control and Estimate at Completion (EAC) forecasting, performance tracking and delay prediction. Adaptive neuro-fuzzy inference system (ANFIS), deep learning, and hybrids anticipate cost variance and assess schedule adherence using earned value metrics and project complexity factors (Dastgheib *et al*., 2022; Uddin *et al*., 2024). CNNs and graph networks support real time anomaly detection. Delay prediction often combines neural models with metaheuristic optimization such as genetic algorithms or particle swarm methods, with reported accuracy above 91% for multi phase risks (Yaseen *et al*., 2020; Bakhshi *et al*., 2022). Graph attention networks further improve performance by modeling task dependencies and inter risk propagation (Mostofi *et al*., 2025a).

4.2 Performance and effectiveness of ML in CPM (RQ2)

Across cost, schedule, and risk tasks, ML consistently improves predictive accuracy and efficiency, with effectiveness tied to robustness, interpretability, scalability, and context.

ANN and ensembles report strong results in cost estimation with R squared greater than 0.95, in delay prediction with accuracy greater than 90%, and in quality classification with values above 96% (Liu *et al*., 2020; Sanni Anibire *et al*., 2022; Fan, 2025). Combining boosting, neural networks, and optimization often improves reliability by 15–50% (Wang *et al*., 2024; Yaseen *et al*., 2020). Early warning horizons for delay risks can reach 299 days in specific settings (Bakhshi *et al*., 2022). Graph based approaches outperform traditional statistics on relational problems such as contingency management and rework cost forecasting

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and can retrain as site data shift (Mostofi *et al*., 2024b). Robustness improves with fuzzy logic and with ensembles in the presence of uncertainty or small samples, and interpretability methods such as SHAP with feature ranking and dashboards enable actionability for PM teams (Dastgheib *et al*., 2022; Kim *et al*., 2022; Chen *et al*., 2024; Elmousalami *et al*., 2025). Despite accuracy gains, adoption is limited by data access, integration hurdles, and compute demands, especially where digital maturity and standardized repositories are lacking (Bang *et al*., 2022; Shahedi *et al*., 2024).

4.3 Implementation success factors and barriers

Effective deployment depends on data foundations, model design and integration, organizational readiness, and ethics.

Studies with richer and better structured datasets often report stronger results, typically with 60 to more than 1,200 records and expert validated features. Data augmentation and dimensionality reduction help when samples are small, including bootstrapping and GAN based synthesis (Liu *et al*., 2020; Chen and Wang, 2024; Khodabakhshian *et al*., 2025; Elmousalami *et al*., 2025). Real time and sensitive data remain difficult to access, and concerns about privacy and ownership persist for personnel and financial metrics (Darko *et al*., 2023).

Models that combine optimization such as dandelion-based optimizer (DBO), particle swarm optimization (PSO), or simulated annealing with neural networks perform well in dynamic scheduling and forecasting but require advanced expertise and compute resources that many small and medium firms do not possess (Egwim *et al*., 2021; Wang *et al*., 2024; ForouzeshNejad *et al*., 2024; Zeng and Gao, 2025). Alignment with existing tools such as Excel, BIM, and Primavera improves usability and reduces resistance, with Excel based interfaces often preferred by practitioners (Assaf and Assaad, 2024; Kusonkhum *et al*., 2022).

Low AI literacy and resistance to change remain common. Fear of displacement and low trust slow adoption. Phased deployment, role specific training, and stakeholder involvement improve outcomes, and early pilots often catalyze broader cultural change (Bang *et al*., 2022; Ding *et al*., 2024; ForouzeshNejad *et al*., 2024; Mostofi *et al*., 2025a).

4.4 Performance patterns from subset synthesis

Analysis of ten comparative studies revealed consistent performance hierarchies across CPM prediction tasks. For cost estimation, hybrid and ensemble methods showed superior accuracy: LightGBM achieved the lowest MAPE of 2.99% on 1,277 projects, DNN-SVR reached 6–7% on 198 projects, and CatBoost with support vector machine (SVM) was especially effective for design change costs. For duration forecasting, tree-based ensembles such RF and GBTs consistently performed best, with the latter 67 times faster than neural networks in direct comparisons. In delay prediction, hybrids combining ML with metaheuristic optimization outperformed single algorithms. Table 2 presents the quality-weighted assessment of each study's methodology and outcomes.

5. Framework development

Building on the review, we propose MLCPM framework shown in Figure 3. It is a multi-layered approach for integrating ML into CPM.

The primary objectives of MLCPM are to (1) provide a structured approach for aligning ML techniques with PMBOK phases Planning, Execution, and Monitoring and Control, and (2) support phase specific adoption that complements construction practice and is usable for both technical and non-technical users.

The framework comprises four interlinked layers: Data, ML Processing, Integration, and Output. The Data Layer underpins ML readiness by organizing five key input types–historical project records, real-time control data, BIM/technical data, stakeholder inputs, and external variables (e.g. weather, supply chain status). These are mapped to appropriate project phases.

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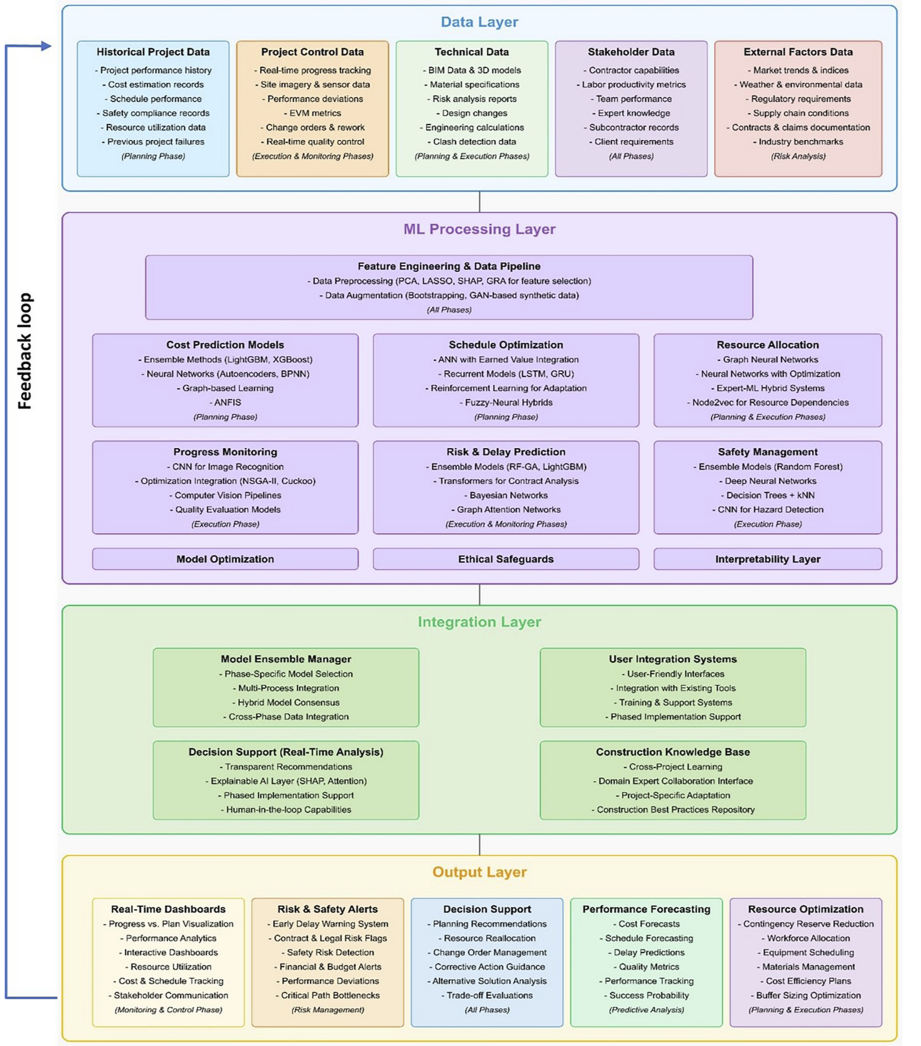


Figure 3. Machine learning–enabled construction project management (MLCPM) framework. Source: Authors' own work

Feature engineering techniques like SHAP, LASSO, and gray relational analysis enhance model input quality, while data augmentation strategies such as bootstrapping and GANs address data scarcity.

The ML Processing layer maps methods to phases. Planning uses XGBoost, LightGBM, and neural fuzzy hybrids for cost scheduling and resources. Execution applies CNN and transformer models for progress safety and contract analytics. Monitoring and control uses ANFIS, LSTM, and deep networks for earned value delay and quality. Interpretability is embedded. A feedback loop tracks metrics such as MAPE and classification accuracy and signals retraining. The Integration layer makes insights usable. An ensemble manager improves reliability. Links to Excel Primavera P6 and BIM ease adoption. Human in the loop practice and role specific interfaces reduce resistance. The Output layer delivers phase specific

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dashboards that include cost forecasts safety alerts delay predictions and quality metrics. Interpretability appears with results. Performance monitoring triggers recalibration when accuracy drifts.

Adoption follows a light path. Begin with expert consultations and an assessment of data assets and systems. Validate on historical projects. Run pilots or sandbox tests against current tools and compare technical and organizational measures.

Safeguards include privacy through anonymization and access control, fairness audits for sensitive decisions, explainable methods, and plans for reskilling. Organizational readiness is treated as a capability that grows during rollout through training and integration with familiar tools. This approach addresses barriers reported in prior studies that include inconsistent data systems and skepticism toward automation (Bang *et al*., 2022; Ding *et al*., 2024).

6. Implications for theory and practice

The synthesis gives clear guidance on model choice. Hybrids and ensembles consistently lead for cost prediction. Tree based ensembles are reliable for duration. Combinations of machine learning with metaheuristic optimization improve delay classification. These results support the multiple model strategy in the MLCPM framework and offer practical starting points across PMBOK phases.

Several implementation priorities follow. First, invest in data foundations including standards preprocessing and governance while addressing the readiness gap in which successful deployments often require data sets from 60 to more than 1,200 projects. Second, evaluate beyond accuracy by emphasizing interpretability such as SHAP, robustness such as fuzzy hybrids, adaptability through retraining triggers, and integration with Excel and BIM. Third, adopt in phases beginning with pilots and expanding as digital maturity grows. Fourth, expect tangible returns with contingency reserves moving from 10 to 15% toward 5 to 8%, schedule prediction improving by more than 30%, resource use improving by up to 30%, and earlier detection of risk.

Broader strategy also matters. Friendly interfaces can widen access, yet the complexity of advanced models such as generative adversarial networks and graph attention networks may require specialization. Large firms can build in-house capabilities while others may partner with external providers. Integrating the framework proposed by Lotfi *et al.* (2024) with predictive workflows can strengthen early-stage risk management and adaptive resource allocation. Advantages will come from smooth integration into daily work and from alignment with emerging data and governance standards that enable cross project learning benchmarking and compliance. Early adopters are likely to benefit as these capabilities scale.

7. Conclusion and outlook

This study systematically reviewed 64 peer-reviewed articles to investigate ML applications in CPM. The findings confirm ML's growing influence across planning, execution, and monitoring phases–enhancing forecasting, decision-making, and control. Yet, implementation gaps remain, particularly in aligning advanced models with real-world construction practices.

To address this, we proposed the MLCPM framework–a theory-driven, phased model that supports ML adoption across varying levels of digital maturity. Grounded in Systems, Complexity, and Dynamic Capabilities Theory, MLCPM reflects construction environments' interconnected, uncertain, and adaptive nature. The framework introduces a four-layered architecture aligned with PMBOK process groups, integrating feedback loops, retraining triggers, and human-machine collaboration. A staged rollout strategy promotes early wins through pilot projects, integration with familiar tools, and role-specific training.

Key findings include:

(1) ML tools have evolved from standalone models to integrated, context-aware systems that enable early risk detection, dynamic scheduling, and real-time resource planning.

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(2) Hybrid and ensemble models–particularly those combining neural networks, fuzzy logic, and optimization algorithms–outperform traditional cost estimation, progress monitoring, and delay prediction methods, often achieving R^2 values above 0.95.

(3) The MLCPM framework provides a structured, theory-informed roadmap for ML deployment in CPM, emphasizing technical accuracy, system integration, and organizational learning.

(4) Phased deployment strategies–starting with small-scale pilots–enhance adoption by embedding ML into familiar workflows and ensuring retraining interpretability and trust-building mechanisms are in place.

(5) Quantitative synthesis showed tree-based ensembles performed best for duration, hybrid ML–metaheuristics for delay, and LightGBM lowest error (2.99% MAPE) for cost.

Limitations of this review include its restriction to English-language, peer-reviewed sources, which may overlook valuable non-academic or regional insights. Inconsistent reporting of model performance metrics limited direct comparisons across studies. Despite technical advances, ethical and legal considerations remain underexplored in the reviewed literature, with few studies addressing algorithmic bias in labor deployment, worker displacement concerns, or liability in automated decision-making (Abbasianjahromi and Aghakarimi, 2023; Sadeghi and Niu, 2024). The performance synthesis covers only 10 of 64 studies due to reporting limitations. Geographic concentration in Asia and Middle East may limit generalizability, and publication bias toward positive results cannot be excluded.

Future research should focus on validating MLCPM through longitudinal case studies in live construction settings. The development of standardized evaluation frameworks will be critical, incorporating both technical metrics and organizational readiness. Additional attention is needed on explainability techniques (e.g. SHAP), ethical auditing, privacy-preserving ML, and human–AI collaboration. Strengthening usability, trust, and scalability will ensure ML delivers meaningful, lasting impact in CPM.

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